

6. [Maximum mark: 8]

Consider the vectors $\mathbf{a} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -8 \\ 9 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} p \\ -6 \end{pmatrix}$, where $p \in \mathbb{R}$.

(a) Find an expression, in terms of p , for

(i) $\mathbf{a} \cdot \mathbf{c}$;

(ii) $\mathbf{b} \cdot \mathbf{c}$.

[3]

The angle between $\mathbf{a}$ and $\mathbf{c}$ is equal to the angle between $\mathbf{b}$ and $\mathbf{c}$.

(b) Find the value of p .

[5]

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